# MILITARY SPECIFICATION

## Projectile, 152MM, Slug, Proofing Metal Parts Assembly

## 1. SCOPE

1.1 This specification is the applicable detail specification for Projectile 152MM, Slug, Proofing, Metal Parts Assembly.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-A-2550 | - Ammunition and Special Weapons, General Specification for |
| MIL-P-10025 | - Packing and Marking for Domestic Shipment of Inert Ammunition Components, General Specification for |
| MIL-R-11073 | - Rotating Band Blanks, Sintered Iron |

STANDARDS

Federal

Federal Test Method Standard No. 151 - Metal; Test Methods

Military

| | |
|---|---|
| MIL-STD-105 | - Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-109 | - Quality Assurance Terms and Definitions |
| MIL-STD-129 | - Marking for Shipment and Storage |
| MIL-STD-639 | - Surface Roughness Values Preferred for Army Ordnance Material |
| MIL-STD-1235 | - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes |

FSC 1320

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DRAWINGS

Munitions Command

| | |
|---|---|
| F10535958 | - Projectile 152MM, Slug-Proofing Metal Parts Assembly |
| F-8889328 | - Pallet for Fixed and Semi-fixed Projectiles |
| IEL-10535958 | - Inspection Equipment List for Projectile 152MM, Slug-Proofing Metal Parts Assembly |

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications and the following:

3.2 First article sample.- Requirements for submission of first article sample by the contractor shall be as specified in the contract.

3.3 Heat treatment or stress relief of Projectile Body.- The heat treatment or stress relief applied shall be uniform throughout each lot of components and shall be so conducted as to produce in any component the mechanical properties specified on the applicable drawings.

3.3.1 Hot forged-heat treat method of manufacture (if applicable).- Projectile body forgings manufactured by this method shall be subjected to heat treatment to attain required mechanical properties. The heat treatment of forgings shall consist of heating above the critical temperature, not to exceed 1600 degrees Fahrenheit (F), followed by an oil quench. After quenching, the piece shall be tempered without delay at a temperature above 700 degrees F., suitable to produce required mechanical properties. After removal from tempering furnace, the projectile forging shall be slowly cooled in air (forced air permitted) to a temperature of not more than 700 degrees F., after which cooling may be accelerated by water spray or similar means.

3.3.2 Cold shaping method of manufacture (if applicable).- Cold shaping of projectile bodies shall be so accomplished as to produce the mechanical properties specified. The projectile body shall be stress relieved at a temperature of 700 degrees F., minimum, following the final forming operation during which time plastic flow takes place. The minimum soak time at or above minimum temperature shall be established which will produce mechanical properties specified by the applicable drawing.



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3.4 Band seat.- Prior to application of rotating band blank, the band seat of the body shall be clean and free of oil, grease, dirt, rust, and other foreign matter. Shot or sandblasting shall not be permitted for cleaning the band seat after knurling.

3.5 Protective coating.- Protective coating shall be in accordance with the applicable drawings and the specifications referenced thereon.

3.6 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.6.1 Metal defects.- All components shall be free from cracks, splits, seams, laps, cold shuts, inclusions or any similar defect.

3.6.2 Burr.- No component shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.6.3 Foreign matter.- All parts and assemblies shall be free of chips, dirt, grease, oil, rust or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality Assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550.

4.2 First article sample.



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4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 15 slug proofing metal parts assemblies as follows:

5 Slug proofing metal parts assemblies, complete, including paint
5 Slug proofing metal parts assemblies, complete, except paint
5 each bodies, prior to banding, including rotating seat
5 each rotating band blanks

The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing shall be the same as used in regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.- After inspection and provisional acceptance at source, the sample shall be inspected for all the requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.- Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-1235, as applicable.

4.3.1.2 Forging lots.- Forging lots shall contain bodies from not more than one heat of steel.

4.3.1.3 Cold shaping lots.- Cold shaping lots shall contain bodies from not more than one heat of steel.

4.3.1.4 Heat treatment lots.- Heat treatment lots shall consist of bodies manufactured from forgings in accordance with 4.3.1.2 that have been heat treated in the same furnace at the same temperature and for the same length of time. After final heat treatment, forgings shall remain grouped for identity purposes until such time as the results of the tension test hereinafter prescribed indicate compliance with mechanical properties of the applicable drawing.

4.3.1.5 Stress relieved lots.- Stress relieved lots shall consist of bodies manufactured from cold shapings in accordance with 4.3.1.3 that have been stress relieved in the same furnace at the same temperature and for the same length of time. After stress relief the cold shapings shall remain grouped for identity purposes until such time as the tension test hereinafter prescribed indicates compliance with mechanical properties requirements of applicable component drawings.



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4.3.1.6 Miscellaneous lots.- Small lots, bodies manufactured from small heats of steel or components that have become separated from previously delivered lots shall at the discretion of the supplier, be grouped into heat treatment or stress relief lots to which a miscellaneous lot number shall be applied. The lot thus constituted (either heat treated or stress relieved) shall form the basis for the selection of samples for the tension test hereinafter prescribed.

4.3.1.7 Projectile lot.- Heat treated or stress relieved lots of components that have complied with the mechanical properties requirements of the applicable component drawing, may be regrouped for assembly into projectile lots. A projectile lot shall consist of projectile assemblies produced by the one manufacturer, under one contract, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL-STD-105 or MIL-STD-1235, as applicable. An AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect shall be used. All nonconforming material shall be rejected. The Ordnance Standard Finish Blocks prescribed by MIL-STD-639 shall be used as a basis of comparison for surface roughness determination.

4.3.2.1 Body - Drawing F10535959 covering a detail of Drawing F10535958.

| | Defect | Method of Inspection |
|---|---|---|
| CRITICAL | | |
| | None defined | |
| MAJOR | | |
| 101. | Concentricity of rotating band seat with bourrelet | Gage |
| 102. | Diameter of rotating band seat | Gage |
| 103. | Location of rotating band seat | Gage |
| 104. | Scoring or knurling missing from rotating band seat | Visual |
| 105. | Metal defective | Visual |
| MINOR | | |
| 201. | Angle on side of rotating band seat | Gage |
| 202. | Flat on side of rotating band seat, max. | Visual (See 6.1) |
| 203. | Radii or chamfers missing or incorrect | Visual (See 6.1) |
| 204. | Surface finish improper | Visual (See 6.1) |
| 205. | Burr | Visual |
| 206. | Foreign matter | Visual |



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4.3.2.2 Metal parts assembly, Drawing F10535958 (before paint).

| Defect | | Method of Inspection |
|---|---|---|
| CRITICAL | | |
| | None defined | |
| MAJOR | | |
| 101. | Concentricity of rotating band with bourrelet | Gage |
| 102. | Perpendicularity of rear face | Gage |
| 103. | Diameter of rotating band | Gage |
| 104. | Diameter, outside, min. | Gage |
| 105. | Length from rear face to forward basic diameter in rear cavity | Gage |
| 106. | Length from rear face to rear basic diameter in rear cavity | Gage |
| 107. | Length from rear face to basic diameter on rotating band | Gage |
| 108. | Length from bottom of rear cavity to face of thread | Gage |
| 109. | Length of effective thread, min. | Gage |
| 110. | Location of rotating band | Gage |
| 111. | Width of rotating band | Gage |
| 112. | Basic angle on rotating band | Gage |
| 113. | Variation of wall thickness | Gage |
| 114. | Metal defective | Visual |
| MINOR | | |
| 201. | Diameter of forward cavity | Gage |
| 202. | Diameter of undercut forward of thread (if applicable) | Gage |
| 203. | Thickness of base | Gage |
| 204. | Width of undercut forward of thread, max. (if applicable) | Gage |
| 205. | Depth of undercut, front and rear of rotating band (if applicable) | Visual (see 6.1) |
| 206. | Width of undercut, front and rear of rotating band (if applicable) | Visual (see 6.1) |
| 207. | Radii or chamfers missing or incorrect | Visual (see 6.1) |
| 208. | Surface finish improper | Visual (see 6.1) |
| 209. | Burr | Visual |
| 210. | Foreign matter | Visual |



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4.3.2.3 Metal Parts Assembly, Drawing F10535958 (after paint).

| Defect | Method of Inspection |
|---|---|
| CRITICAL | |
| None defined | |
| MAJOR | |
| 101. Weight | Approved scale |
| 102. Pitch diameter of thread, min. | Gage |
| 103. Major diameter of thread, min. | Gage |
| 104. Diameter, outside, max. | Gage |
| 105. Thread(s) damaged or not full | Visual |
| MINOR | |
| 201. Pitch diameter of thread, max. | Gage |
| 202. Paint on threads | Visual |
| 203. Paint not dry | Visual |
| 204. Protective coating inadequate or damaged | Visual |
| 205. Marking missing, incorrect or illegible | Visual |
| 206. Burr | Visual |
| 207. Foreign matter | Visual |

4.3.2.4 Pallet, Drawing F8889328.

| | |
|---|---|
| CRITICAL | |
| None defined | |
| MAJOR | |
| None defined | |
| MINOR | |
| 201. Pallet damaged (unserviceable) | Visual |
| 202. Sleeve(s) missing | Visual |
| 203. Improperly packed | Visual |
| 204. Strapping missing, loose or broken | Visual/Manual |
| 205. Marking missing, incorrect or illegible | Visual |

4.3.3 Inspection testing.- The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Mechanical properties.



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4.3.3.1.1 Hardness.

4.3.3.1.1.1 Forging lot. (not requiring heat treatment) - Each body in the forging lot (as defined by 4.3.1.2) shall be subjected to Rockwell Hardness test using the "C" scale to determine the bodies to be selected for the tension test.

4.3.3.1.1.2 Heat treatment lots.- In order to determine uniformity of heat treatment, each body in the heat treatment lot (as defined by 4.3.1.4) shall be subjected to a Rockwell Hardness Test using the "C" scale, to determine the bodies to be selected for the tension test.

4.3.3.1.1.3 Stress relieved lots.- Twenty-five bodies from the first batch processed through the stress relief furnace from each stress relieved lot (as defined by 4.3.1.5) shall be subjected to a Rockwell hardness test using the "C" scale to assess uniformity of the product.

4.3.3.1.1.4 Miscellaneous lots.- Miscellaneous heat treatment lots of bodies shall be hardness tested in accordance with 4.3.3.1.1.1. Miscellaneous cold shaping bodies shall be hardness tested in accordance with 4.3.3.1.1.2.

4.3.3.1.2 Tension test.

4.3.3.1.2.1 Forging lot (not requiring heat treatment).- Two body forgings, one of the hardest and one of softest (as determined by 4.3.3.1.1.1) shall be subjected to this test. Failure of any tension test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the lot. At the option of the supplier, a production control sample may be utilized to establish a hardness range. When a production control sample is used, the hardness range thus established shall be maintained throughout the forging lot. The size of the sample shall be determined by the contractor. Slug bodies which fall outside of the established range may be heat treated to fall within range or new mechanical properties may be determined from newly selected projectile bodies which will represent the true extremes of hardness in the forging lot.

4.3.3.1.2.2 Heat treatment lots.- Two body forgings, one of the hardest and one of the softest (as determined by 4.3.3.1.1.2) shall be subjected to this test. Failure of any tension test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the lot. At the option of the supplier, a production control sample may be utilized to establish a hardness range. When a production control sample is used, the hardness range thus established shall be maintained throughout the heat treatment lot. The size of the sample shall be determined by the contractor. Bodies which fall outside of the established range may be re-heat treated to fall within the range or new mechanical properties may be determined from newly selected bodies which will represent the true extremes of hardness in the heat treatment lot.



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4.3.3.1.2.3 Stress relieved lots.- Two bodies from each stress relieved lot shall be subjected to this test to determine compliance with the mechanical properties requirements of the applicable drawing. Failure of one or more tension test specimens to comply with the yield strength or elongation requirements shall be cause for rejection of the lot.

4.3.3.1.2.4 Miscellaneous lots.- Samples for tension test for miscellaneous heat treatment or stress relieved lots shall be selected and tested in accordance with 4.3.3.1.2.1, 4.3.3.1.2.2 or 4.3.3.1.2.3 as applicable.

4.3.3.2 Rotating band.

4.3.3.2.1 Band seating.- Two body and band assemblies from each lot, prior to painting, shall be subjected to this test to determine compliance with the applicable drawing. Failure of either assembly to comply shall be cause for rejection of the lot.

4.3.3.2.2 Density.- The sample provided by 4.3.3.3.1 (after examination for seating) shall be subjected to this test to determine compliance with the requirements of the applicable drawing. Failure to comply shall be cause for rejection of the lot.

4.3.3.3 Cleaning and application of protective coating.- The sample size and acceptance criteria shall be as specified in the applicable specification. Failure to comply with the requirements shall be cause for taking the corrective action prescribed therein.

4.3.4 Inspection equipment.- Index of Inspection Equipment List IEL-10535958 identifies the inspection equipment required to perform the examination and tests prescribed herein. Equipment design shall be in accordance with code designation specified on the applicable IEL.

4.4 Test method and procedures.

4.4.1 Mechanical properties.

4.4.1.1 Hardness.- After final heat treatment (if applicable) or stress relief (if applicable) and before final machining, the hardness test shall be performed on the body at the rear bourrelet forward of rotating band seat. The tests shall be made using the procedure for performing this test as prescribed by Federal Test Method Standard No. 151.

4.4.1.2 Tension test.

4.4.1.2.1 Yield strength and elongation.- From the bodies selected in accordance with 4.3.3.1.2.1, 4.3.3.1.2.2, 4.3.3.1.2.3 and 4.3.3.1.2.4, two tension test specimens shall be taken longitudinally from mid point in the body 180 degrees apart. The specimens shall conform to the largest obtainable standard "R" type specimen. All specimen types shall conform to Federal Test Method Standard No. 151. Yield strength shall be determined by the offset method utilizing the procedures of Federal Test Method Standard No. 151.



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4.4.2 Rotating band.

4.4.2.1 Band seating.- From the sample provided by 4.3.3.2, the body and band assembly shall be carefully sectionalized with a wet abrasive cut-off wheel, so as to produce four segments of 90 degrees each. Visual examination of the exposed surfaces of the rotating band shall then be conducted to determine compliance with the applicable drawing.

4.4.2.2 Density.- After examination for band seating as prescribed in 4.4.2.1, four specimens, spaced approximately 90° apart, shall be removed from the rotating band. The specimens shall be the full width and thickness of the band and approximately one inch long. The specimens shall be sawed to the specified size and the burrs removed. The wax shall be removed by either of the following methods:

a. Place specimens in a dry hydrogen atmosphere furnace and hold at 1400° F for one hour. Remove specimens and weigh. Return specimens to the furnace and hold at 1400° F for 30 minutes. This procedure with the half hour heating intervals shall be repeated until the change in weight is less than .05%.

b. A Soxhlet apparatus of suitable size using either toluene or trichloroethylene may be used for extracting the wax. The extraction time shall be a minimum of 18 hours. The specimen shall then be dried for 1½ hours in an oven at 210° F. This procedure with 1½ hour extraction intervals shall be repeated until the change in weight is less than .05%.

Following wax removal and weighing, the specimens shall be impregnated with micro-crystalline wax. This shall be accomplished by reducing the pressure over the specimen immersed in molten micro-crystalline wax held at 180° to 200° F to not more than 2 inches of mercury pressure for 30 minutes by a suitable evacuating method. The pressure shall then be slowly increased (approximately 30 seconds) to atmospheric pressure after which the specimen shall remain immersed in the molten micro-crystalline wax at 180° to 200° F and at atmospheric pressure for approximately 1 hour.

The density of the specimens shall be calculated as follows:

$$D = \frac{A}{B-C}$$

Where

D = density in grams per cubic centimeter
A = weight in air of wax free specimen, in grams
B = weight in air of the micro-crystalline impregnated specimen, in grams
C = weight of micro-crystalline wax impregnated specimen immersed in water, in grams



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4.4.2.3 Density of band blank.- The procedure for determining density shall be as specified in 4.4.2.2 except than nine specimens shall be selected from the band blanks selected in accordance with 4.3.3 of MIL-R-11073 as indicated in fig. 1 to determine compliance with applicable drawing. Three sets of specimens as indicated on fig. 1 spaced approximately 120° apart shall be sectioned from each blank. The size of each specimen shall be approximately one inch long X approximately 1/3 the width of the band X the full thickness of the band blank.

4.4.3 Protective coating.- The procedures for testing the cleaning of components and the application of protective coating shall be in accordance with the criteria for testing as specified in the applicable specification.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 Level A - Military Pack - Not applicable.

5.1.2 Level B - Limited Military Pack - Not applicable.

5.1.3 Level C - Minimum Military Pack -

5.2 Packaging and packing.- Packaging, packing and marking shall be in accordance with MIL-P-10025.

5.2.1 Slug, Proofing, Metal Parts Assembly shall be palletized in accordance with Drawing F8889328.

5.3 Data Card.- Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 Visual examination qualification.- When compliance with the applicable requirements is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

6.2 Order data.- Procurement documents should specify the title, number and date of this specification.

Custodian:

Army - MU

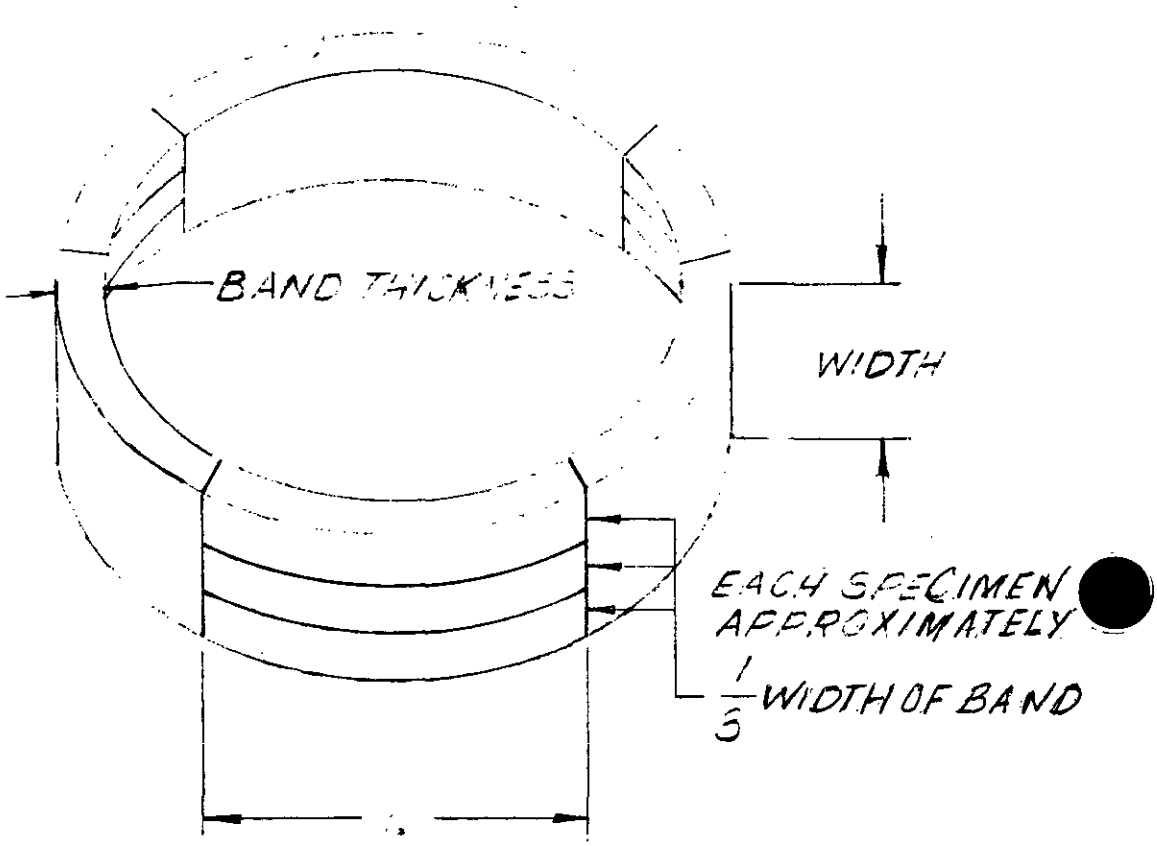
Preparing activity:

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Project No. 1320-A097



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THREE SETS OF SPECIMENS AS INDICATED. SHALL BE SECTIONED FROM EACH BAND TO BE TESTED. SPACED APPROXIMATELY 120° APART.

FIG 1



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

| ORGANIZATION | CITY AND STATE |
|---|---|
| | |

| CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT |
|---|---|---|
| | | $ |

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

| SUBMITTED BY *(Printed or typed name and activity)* | DATE |
|---|---|
| | |

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